

Google's Astronomical Influence Campaign

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Abstract

In an uncertain funding landscape, corporate philanthropy and partnerships are of greater import to scientists. One such industry-academic relationship between Google and astronomy demonstrates how such partnerships might be complicated by commercial interest in the field of study. This essay seeks a comprehensive view of Google's involvement in astronomy, and what forces motivate such involvement. Using contemporary journalism, I will describe Google's commercial interests in the space industry, and how that affects the specific means by which Google interacts with astronomers, namely through computational resources and direct funding. I assess the extent of astronomer's dependence on Google's software architecture in part via an independent study of the propensity of Google-built software in astronomy publications. I find that, taken as a whole, Google seems to be maneuvering towards various means of control in astronomy, perhaps to safeguard the company's initiatives in space from astronomers' criticism. Finally, I suggest that astronomers must move towards active forms of political expression, should Google and Big Tech's commercialization, and ergo pollution, of the night sky be resisted.

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Introduction

“For better or worse, the practice of science in the 21st century is becoming shaped less by national priorities or by peer-review groups and more by the particular preferences of individuals with huge amounts of money.” Steven A. Edwards, American Association for the Advancement of Science.¹

The modern American scientist finds themselves in a landscape wrought with precarity. They realistically have two options, especially if they do research without immediate commercial applications. One is to apply for dwindling federal funding, more likely than not from the Department of War,² with the chance of weeks-long delays even if an application is accepted.³ The other is to turn to private benefactors, pitching research projects to the world’s billionaires with extra cash to spend. In the new regime of data capital, more often than not these are those who obtained their wealth in tech.⁴ But what does one make of, for example, the multimillion dollar donations to science sent by Google⁵ while it simultaneously funds the far right,⁶ which continues to push for decreases in federal funding to science? How are both actions beneficial to the same company?

There is perhaps no field in which these two options are more mandatory than astronomy. Here, one must either secure funding through various governmental agencies—NASA or the NSF, typically—or find a position in the billionaire-funded research institutes of Simons, Kavli, or Moore. Additionally, there is perhaps no scientific field more crucial to tech companies’ interests than astronomy, besides the obvious computer science and possibly fundamental physics. Of such corporations involved in the so-called billionaire space race, Google’s strategy

¹ Broad, “Billionaires With Big Ideas Are Privatizing American Science.”

² Blevins et al., *Federal Research and Development (R&D) Funding: FY2026*.

³ E.g. Kozlov et al., “White House Stalls Release of Approved US Science Budgets.”

⁴ Helman et al., “America’s Most Generous Philanthropists 2025.”

⁵ Google Org Impact Challenge, “Google.Org Impact Challenge.”

⁶ Paoli and Editors, “Meet All 37 White House Ballroom Donors Funding the \$400 Million Build, Including Silicon Valley Tech Giants, Crypto Bros and the Lutnicks.”

is unique in its marked involvement with astronomers, frustrating them by funding fleets of telescope-blocking satellites, but also supporting them in research collaborations and funding opportunities.

This project will analyze Google's influence on astronomical research as a case study of Google's broader influence on scientific knowledge production. Such a case study is motivated by three reasons. Google has ample commercial interest in the space sector, and therefore in space professionals: engineers, physicists, and astronomers whose research careers are intertwined with the development of space technologies. Google is omnipresent in modern work environments, increasingly so in astronomical research projects, primarily through Google-developed open-source software and data storage in the cloud. Google has significant precedent of influencing similar professionals, namely academics of other fields, towards their commercial interests.⁷ As such, Google's involvement in astronomical research stands to grow in proportion to the company's involvement in the space industry—an involvement that, as I will describe below, is increasingly mandatory for astronomers.

Astronomy in the 21st Century

Astronomy is one of the oldest sciences studied by humankind, dating back to over 2 millennia BCE.⁸ Most generally, astronomy is the study of everything beyond the Earth's atmosphere, such as the Sun, planets, asteroids, galaxies, and in the modern era, man-made satellites. In many universities, astronomy research exists under various departments: physics, astrophysics, space science, planetary science, and so on are home to astronomy researchers studying various subfields. This work thus considers all research related to outer space as a form

⁷ E.g., Zuboff, *The Age of Surveillance Capitalism*, 86–87.

⁸ Friedlander, "History of Astronomy."

of astronomy—referencing physicists, rocket engineers, and computer scientists alike should their research pertain to outer space.

Like humanity itself, the field has changed alongside technological advancement, most notably by the advent of the telescope, the computer, and rocketry capable of escaping Earth's gravity. The state of the field today is defined by these three technologies. The world now hosts over 25 observatory systems,⁹ which include hundreds of ground-based telescopes. There are ~30 telescopes orbiting in space,¹⁰ and this number is increasing as more nations obtain launch capability. These telescopes on the ground and in space produce huge amounts of data, which require the use of advanced computing tools to parse. Such computing power has been similarly utilized for theoretical purposes: as telescopes uncover an increasingly complex cosmic system, theorists turn to high-performance computers to simulate various models of the universe.

Research in machine learning and artificial intelligence (ML/AI) have opened new pathways for astronomers to analyze data, observed or simulated. Indeed, as the volume of data only stands to increase,¹¹ such technologies are essentially mandatory to parse through data efficiently. It has furthermore become the norm to release open-source software online, typically on the platform Github, a subsidiary of Microsoft since 2018.¹² Open-source ML/AI software have primarily developed out of the research divisions of tech corporations. In the programming language Python, one of the most common languages used in astronomy, open-source ML/AI software primarily stems from Google (and its subsidiary DeepMind) and Meta (through its Meta AI research group). Google, due to the company's immense cloud storage resources and open source software, is particularly influential in astronomy.

⁹ Editors, *List of Astronomical Observatories and Telescopes*.

¹⁰ Li, "An Overview of Active Space Telescopes."

¹¹ E.g. "Data | Rubin Observatory."

¹² Cunningham, "GitHub Will Be Folded into Microsoft Proper as CEO Steps Down."

Astronomy has historically been funded by wealthy individuals,¹³ however, the conditions of such funding have changed—now, benefactors have material interests in space. That is, echoing the commercialization of geological and environmental research following the expansion of the fossil fuels industry,¹⁴ it is possible that astronomy will commercialize as the space industry expands. Akin to pharmaceutical or oil funding, Google appears to enter as a private funder set to change how knowledge in astronomy is produced.

In contrast to prior corporate precedent in geoscience, Google's influence in astronomy is not because Google is a space company. While Google supports a network of astronomical projects, it likewise has partnerships in bioengineering, climate science, and more through the company's massive computing resources.¹⁵ Through the widespread necessity of computing, Google has become an everything company. Unlike previous precedent of corporate influence in academic research, Google's commercial aims reach comprehensively across essentially all fields of science. However, it seems that astronomy is of particular import as, while outer space is not the center of Google's work, it is the next venue in Google's quest for computational hegemony.

Precedent in Fossil Fuels

Literature on the fossil fuels industry influence on academia presents a grounded example of these forces at play. Fossil fuel conglomerates seeking to leverage the social and governmental influence of the trusted academic expert class¹⁶—in other words, to secure cultural hegemony—have ingratiated themselves into the modern university, funding research centers,

¹³ "The Funding Frontier."

¹⁴ Yoder, "How Big Oil's Big Money Influences Climate Research."

¹⁵ Google Org Impact Challenge, "Google.Org Impact Challenge."

¹⁶ Tyson and Kennedy, "Public Trust in Scientists and Views on Their Role in Policymaking."

conferences, and student services.¹⁷ Such partnerships resulted in material gains, such as favorable publications and faculty features in fossil fuels advertisements,¹⁸ but an additional opportunity to platform the underpinning right-wing ideology of fossil fuels corporations. This ideology is curated to preserve the status quo, featuring “the propagation of fear and a discourse of security, the notion of inevitable technological innovation, [and] growth and maintenance of current patterns of consumption,” among others.¹⁹

Google’s academic influence in the university takes on similar themes as those in fossil fuels, however, deviates from such precedent due to the company’s all-encompassing nature. Google has already captured a significant market share in modern working life, lending an implicit hegemony. Google similarly courts academics to expand its cultural sphere²⁰ deeper into the expert class, with research collaborations fulfilling a similar role of normalizing the company and its ideologies. Indeed, the very same ideological facets of the fossil fuels industry likewise appears for Google: expanded collaboration with the federal government to maintain national security²¹ and the unfettered growth of the AI industry²² despite the wide-ranging social and environmental impacts.²³ With these ideological lines lies the implicit assumption of technological progress as a force of nature unguided by human interest, that is, the maxim that if Google doesn’t win the AI race, someone else will.

This work aims to contribute to the preexisting literature by surveying a nascent, yet immensely profitable, industry: outer space. As the oil boom fundamentally changed the face of geoscience, the commercialization of outer space will forever change the field of astronomy.

¹⁷ Hiltner et al., “Fossil Fuel Industry Influence in Higher Education.”

¹⁸ Ladd, “Priming the Well.”

¹⁹ Gustafson, *Fossil Knowledge Networks: Industry Strategy, Public Culture and the Challenge for Critical Research*, 315.

²⁰ E.g. <https://www.wsj.com/articles/paying-professors-inside-googles-academic-influence-campaign-1499785286>

²¹ E.g. <https://www.nytimes.com/2025/08/04/technology/google-meta-openai-military-war.html>

²² theconversation.com/google-is-going-all-in-on-ai-its-part-of-a-troubling-trend-in-big-tech-257563

²³ <https://www.unep.org/news-and-stories/story/ai-has-environmental-problem-heres-what-world-can-do-about>[https://](https://www.unep.org/news-and-stories/story/ai-has-environmental-problem-heres-what-world-can-do-about)

While the technologies to exploit natural resources in outer space are yet to reach markets for some decades, a forward-looking review may identify potential pitfalls to guide future policy.

Literature Review

Much of the available writing on Google's influence in the university comes from journalism; there are comparatively few scholarly sources. There are several sources on the use of Google's services as a pedagogical tools and on student perceptions of Google, however, these subjects are not in the scope of this work and are therefore ignored. Below is a brief overview of writing on the influence of Google's services, primarily the search engine, on academic work.

In her 2007 book *The University of Google: Education in the (Post) Information Age*, Tara Brabazon details the effects Google has on a population in which digital literacy was still a nascent concept, particularly for university students. Not unlike the contemporary discourse on AI in education, Brabazon bemoans that “Google is facilitating laziness, poor scholarship and compliant thinking,”²⁴ via its search engine “facilitat[ing] a quick and simple ‘method’ for completing student assignments,”²⁵ resulting in poorly thought-out essays with unreliable sources. Brabazon extends this critique to broader society by noting the influence Google search has on what information people see first—or at all—and, by extension, the company's influence on politics and culture. Brabazon comments on the 2004 implementation of Google Scholar as a valuable solution, but cites its reference to paywalled sources as an accessibility issue.²⁶

In 2009, Siva Vaidhyanathan made similar arguments in an Almanac of Higher Education piece, “The Googlization of Universities.” In it, Vaidhyanathan describes a shared culture between academia and Google—after all, Google was created by two Stanford PhD

²⁴ Brabazon, *The University of Google: Education in the (Post) Information Age*, 15.

²⁵ Brabarzon, 30.

²⁶ Barbrazon, 45.

candidates—facilitating a tightening relationship. Similar to Barbrazon, Vaidhyanathan describes the use of search engines to produce unreliable online sources cited in student work.²⁷

Vaidhyanathan also comments on Google Scholar, and further problematizes the technology for its opaque and imperfect source ranking system.²⁸ In another analog to AI, Vaidhyanathan notes the issues of copyright infringement on the part of Google as the company rapidly digitized books, including those under copyright. Lastly, Vaidhyanathan raises concern in academics' use of Google's advanced computers and data storage, and the possibility of aggregating enough data to find correlation anywhere.²⁹

Another work is Jamali & Asandi (2010), which coincidentally surveyed physics and astronomy academics to find a growing reliance on Google as a search engine for academic articles.³⁰ They raise concerns about the growing use of Google over traditional library services, flagging that “some scientists now consider that what is not online is not worth the effort to obtain.”³¹ Following the above works was a relative lull in publications about Google's impact on universities as a system, perhaps due to the populous' increased familiarity with search engines and the rise of digital literacy campaigns.

References to Google's influence on universities resurfaces in scholarly work in Shohana Zuboff's 2019 book, *The Age of Surveillance Capitalism*. Here, Zuboff describes Google's recruitment of academics to their case against antitrust regulations. Zuboff outlines particular strategies, including the secret commission of pro-Google conferences in 2012 and 2014, direct funding to academics who shared Google's position on public policy from 2009 to 2017, and the ousting of scholars critical of Google from a think tank funded by Google CEO Eric Schmidt.³²

²⁷ Vaidhyanathan, *The Googlization of Universities*, 68.

²⁸ Vaidhyanathan 69-70.

²⁹ Vaidhyanathan 69-72.

³⁰ Jamali and Asadi, “Google and the Scholar.”

³¹ Jamali and Asadi, 291.

³² Zuboff, *The Age of Surveillance Capitalism*, 86–87.

Most recently, there is Sarfi et al. (2021),³³ which examines Google's influence on the university in terms of a soft power propaganda campaign. They first describe the ways in which Google services are ingrained in modern working life, and ergo the academy. Similar to Zuboff, Sarfi et al. trace Google's systematic payment or suppression of academics, and comment on Google's rhetoric regarding its relationship with universities—that is, one which deflects any benefit to Google and emphasizes the good Google does for academia. In addition, Sarfi et al. comments on Google's philanthropic donations towards related student programs, such as cybersecurity. Uniquely, Sarfi et al. (2021) employs two theoretical concepts, Gramsci's *cultural hegemony* and Althusser's *interpellation* to frame Google's intentions in, and influence of, the academy. This same theoretical scaffolding is used in this work, and described in more detail in *Theoretical Grounding*.

This work will contribute to the literature in the following ways. One, while Vaidhyanathan notes the issues of veracity in the scale of data that academics are able to store through Google, and Sarfi et al. (2021) describe the financial influence Google has by funding research groups and programs studying computing, neither describe, as this work will, Google's provision of computing resources as, in and of itself, a persuasive force for academics. Secondly, I present examples of corporate influence in academia from fossil fuels, predating, yet echoed within Google's influence initiatives. Thirdly, the literature referencing specific academics in Google's influence campaign typically note those in computer science, law, or public policy, the latter often discussed more thoroughly than the former, as in Zuboff (2019) and Sarfi et al. (2021). This work will extend the precedent and frameworks set in Zuboff (2019) and Sarfi et al. (2021), and apply it to a field-specific case study of astronomy. In doing so, this work will

³³ Sarfi et al., "Google's University?"

further arguments regarding Google's ubiquity in modern working life to the company's ubiquity in computation-heavy astronomy research, which occurs in avenues distinct from those listed in Sarfi et al. (2021).

Research Question

In what ways are Google involved in the field of astronomical research, and what is Google's motivations behind such involvement?

Methodology

I pull from primary sources—newspaper articles, press releases, statements—to map Google's commercial interests in the space industry, the use of Google's services in astronomy, and Google-related philanthropy to astronomy. I conduct original research, documenting the propensity of Google-written softwares JAX and TensorFlow in astronomy publications through keyword searches on arXiv and SciX, popular online repositories of scientific research prior to and following publication, respectively. I document the use of one such software, JAX, in astronomy through the number of astronomy-specific codes based on it, acquired from an aggregated list on Github. I take the theoretical frameworks of cultural hegemony and interpellation from Sarfi et al. (2021) to analyze how these pieces of evidence demonstrate a form of influence on astronomers.

The above methodology was selected to touch on the two sides of astronomy, observation and theory, thoroughly. Astronomical observing is documented much more often in popular media, a function of observing's capability to produce attractive visuals. Therefore, newspaper articles and online statements by astronomy organizations are sufficient to provide ample evidence to draw from. Theoretical astronomy—or more accurately, theoretical astronomy and the parts of observational astronomy that require intensive computing—is much less attractive to

typical readers, and is therefore less reported. For this reason, it is necessary to pull on software documentation and published journal articles to quantify Google's impact on such endeavors.

Theoretical Grounding

Drawing from the Sarfi et al. (2021) analysis of Google's academic influence on universities more broadly, this work will employ the Marxist concepts of cultural hegemony and interpellation.

Cultural hegemony describes the process by which ruling powers actively mold society by influencing cultural institutions in order to sustain their social dominance.³⁴ In this work, ruling powers refers to the tech billionaire class typified by Google executives, and cultural institutions refers to the knowledge-production centers of astronomy: the university, observatories, and laboratories. As argued in this work, Google has commercial interests in maintaining an image of benevolence, and seeks to sustain such an image via its contributions to science broadly and, due to the company's aims to commercialize space, astronomy in particular.

Interpellation, in turn, is the way in which individuals are normalized to the ruling system.³⁵ Interpellation is the motive force of cultural hegemony; if cultural hegemony is the goal of a ruling class, interpellation is the way to get there. In this work, interpellation appears in the relationships between Google and astronomers. Through philanthropic partnerships with astronomers, Google normalizes its own existence, an existence predicated on the sale of private data³⁶ and flagrant violation of antitrust law.³⁷ The ideological underpinnings of such a system—that is, that it is just or necessary—are thus further entrenched in astronomy alongside Google itself.

³⁴ Gramsci, *Selections from the Prison Notebooks*.

³⁵ Althusser, "Ideology and Ideological State Apparatuses."

³⁶ AP News, "Google Facing \$425.7 Million in Damages for Nearly a Decade of Improper Smartphone Snooping."

³⁷ <https://www.justice.gov/opa/pr/departments-justice-wins-significant-remedies-against-google>

An Overview of Findings

In this work, Google's influence on astronomers is organized into three main themes: *profitability*, *ubiquity*, and *philanthropy*. These three themes are overlapping and inexact; however, the distinction is valuable insofar as each category traces a unique body of literature. *Profitability* follows a lineage of literature on military-astronomy relations, with large corporations such as Google occupying a similar role that the US military once did. *Ubiquity*, with less one-to-one historical precedent, leans the most on Sarfi et al's description of interpellation. *Philanthropy* draws from prior literature from critical university studies, with Google affecting astronomers in similar ways to corporate actors with universities.

Profitability includes Google's commercial interests in the industry surrounding astronomy. These commercial interests largely relate to internet satellites, an industry that Google is keen on as a web-based corporation and actively invests in through the company's venture capital wing, Google Ventures. Satellites are where Google impacts astronomy most directly, with the growing number of satellites in orbit regularly disrupting ground-based telescopes.³⁸ While satellite manufacturers and astronomers often meet to minimize this disruption, plans for thousands-strong satellite megaconstellations are still underway, to the frustration of many in the field.³⁹ Google's commercial plans for astronomy also contain "moonshot" concepts, such as launching AI data centers into space.⁴⁰ Similar critiques have been levied against such concepts for their potential impact on ground-based observing or the environment of outer space (e.g. space debris).

³⁸ Waldek, "Scientists Analyze 76 Million Radio Telescope Images, Find Starlink Satellite Interference 'Where No Signals Are Supposed to Be Present.'"

³⁹ Tonkin and Massey, "SpaceX and Reflect Orbital Plans Would 'permanently Scar' Night Sky."

⁴⁰ Whitwam, "Meet Project Suncatcher, Google's Plan to Put AI Data Centers in Space."

Ubiquity describes the system by which Google essentializes itself to modern working life via the company's myriad online services. Indeed, it is difficult, if not impossible, to work in the modern, digitized world without using one of Google's services—Google is ubiquitous. Such widespread use is echoed in astronomical research. Google's common services—Drive, Gmail, Search, and so on—are naturally of utility to astronomers; however, Google's AI agent Gemini and Cloud computing services have additionally been leveraged for research projects for AI and data storage capacity, respectively. Google also develops JAX and TensorFlow, two codebases which support the programming of machine learning and artificial intelligence (ML/AI) tools and accelerate compilation speeds. Such tools are essential for the specialized programs and complex models written to parse through the large volume of data that observatories put out, and their use is on the rise in astronomy. In fact, many scientists, including astronomers, have transitioned from scientific research to developing JAX and TensorFlow at Google, in support of scientific research.

Philanthropy encompasses Google's charitable activities—in the scope of this work, primarily Google's financial support for scholars. These activities may serve Google's interests directly (e.g. donating to research initiatives arguing against antitrust legislation) or indirectly (i.e. improving Google's public image). The subtextual threat of withholding such funds is also a valuable coercive tool to ensure that recipient organizations maintain good terms with the company. In at least one case, the threat of losing funding directly led to an organization's firing of a research team critical of Google.⁴¹ Thus, regardless of whether such donations are recurring or one-time, it is in the interest of the recipient organization to remain in Google's good graces for fear of losing future funding. In this essay, I use Google's \$1 million donation to Lick

⁴¹ Vogel, "Google Critic Ousted From Think Tank Funded by the Tech Giant."

Observatory and the fully-funded construction of an entire observatory system by ex-Google CEO Eric Schmidt as examples of such a process occurring in astronomy.

Google and Space Profitability

Google's forays into space commercialization regularly clash with the priorities of astronomers. Indeed, space commercialization writ large has faced criticism from the astronomy community, including ventures such as private satellite fleets,⁴² space tourism,⁴³ and AI datacenters in space.⁴⁴ These projects leave bright trails in the night sky, polluting dark sky areas and blocking portions of the sky from ground-based observatories.⁴⁵ They also contribute to the production of orbiting space junk hazardous to future missions, which may curtail all future space exploration in a process known as Kessler syndrome.⁴⁶ As commercial satellite projects rapidly increase in number and scale, impacts on astronomical research, not to mention on nocturnal wildlife and dark sky areas, are expected to rise. Google's involvement in space commercialization exists primarily through its parent company, Alphabet's ~7% ownership of SpaceX, the second-largest proportion behind only Elon Musk.⁴⁷

The Military Precedent

Satellite telecommunications fleets were among the first technologies to significantly mar astronomical research. However, these projects were not primarily advanced through corporate actors as they are today. Predating the commercial space industry, the US military was the predominant force launching material into space. A 1960s US Air Force project to unleash two

⁴² DarkSky International, "DarkSky International Opposes Reflect Orbital's Proposed Orbital Illumination System."

⁴³ Whipple, "Space Tourism Threatens Tranquillity of Moon, Scientists Fear."

⁴⁴ Tonkin, "SpaceX and Reflect Orbital Plans Would 'permanently Scar' Night Sky."

⁴⁵ Witze, "SpaceX Launch Highlights Threat to Astronomy from 'Megaconstellations.'"

⁴⁶ ESA, "The Current State of Space Debris."

⁴⁷ KeepTrack, "Who Owns SpaceX in 2026?"

fleets of copper needles in polar and equatorial orbits around Earth to further long-range radio communication, named Project West Ford, faced mass objections from astronomers. Such a project raised concerns over a potential norm of telecommunication needle fleets across the globe, precipitating a sharp increase in background noise pollution in the sky. Following direct appeals to the project, operated out of the Massachusetts Institute of Technology's Lincoln Laboratory, resolutions against such a project from the International Astronomical Union (IAU), American Astronomy Society (AAS), and similar organizations, and outrage to the press, West Ford was revised to ensure that the needles would be a temporary fixure—they would be engineered to succumb to the Sun, friction, and gravity within a few years. Sure enough, the effects to astronomy were minimal and the vast majority of needles fell from the sky within the designated time.⁴⁸ I begin with Project West Ford as a case study where nearly everything goes right: a US government project raises concerns for astronomy, and astronomers follow typical channels of advocacy to achieve a compromise. This is no small feat when pressing the US military, and in the end the impact was minimal regardless. However, we now live in a new regime, ruled by private enterprise with few channels to the top.

Why is Google in Space?

Internet access is crucial to Google's business. Online advertisements, which accounted for 75.6% of Google's 2024 revenues,⁴⁹ require real-time analysis, with metrics typically relayed to clients at an hourly rate.⁵⁰ It is thus natural that Google takes immense interest in the expansion of internet access, where space-based satellites take a leading role. With more people on the internet, Google obtains more people to advertise to.

⁴⁸ Rand, Lisa Ruth. "From Sputnik to Starlink: historical perspectives on astronomy and satellite communications." *Proceedings of the International Astronomical Union* 20, no. S385 (2024): 5-10.

⁴⁹ "Alphabet Inc. FORM 10-K."

⁵⁰ "About Data Freshness - Google Ads Help."

It is with this perspective that the following sections explore Google's satellite telecommunication endeavors and impact on astronomy. I contend that satellite internet services such as SpaceX's Starlink, of which Google is a major stakeholder,⁵¹ are not thoroughly critiqued, especially in astronomy circles. SpaceX puts forth the prevalent narrative that, while such satellites are detrimental to astronomy, they are ultimately beneficial to the world in that they provide internet access to otherwise inaccessible areas.^{52,53} While this is indeed a boon of these technologies, what is often overlooked is that first and foremost, satellite internet is beneficial to Google and similar digital monopolies. Satellite internet services open a previously untapped market for such corporations to track⁵⁴ and advertise to, not to mention the expanded prospects of government surveillance.⁵⁵ Google, in particular, has immense material interest to expand an internet run on a search engine market that Google controls 90% of.⁵⁶ As such, the paradigm is flipped: tech corporations, including Google, do not fund and operate services such as global telecommunications for the benefit of the world with a bonus of making profit on the side. It is rather a Trojan Horse, in which Google stands to make massive profits off of the expanded market from increased internet access, packaged as internet access for all. Indeed, Starlink is not Google's first foray into satellite internet. The company invested over \$1 billion in similar satellite internet projects in 2014,⁵⁷ previously interfered with the Square Kilometer Array (SKA) radio telescope with internet balloons,⁵⁸ and maintains investments in other satellite

⁵¹ Winkler et al., "Google Nears \$1 Billion Investment in SpaceX."

⁵² Bjarin, "Elon Musk's Vision For Starlink: A Global Communications Revolution."

⁵³ Starlink, "Starlink Stories."

⁵⁴ AP News, "Google Facing \$425.7 Million in Damages for Nearly a Decade of Improper Smartphone Snooping."

⁵⁵ Woollacott, "US Government Demands Millions Of Google, Apple And Meta User Accounts."

⁵⁶ StatCounter Global Stats, "Search Engine Market Share Worldwide."

⁵⁷ Palermo, "Google Invests Billions on Satellites to Expand Internet Access."

⁵⁸ Chirgwin, "Google Launches Broadband Balloons, Radio Astronomy Frets."

internet providers.⁵⁹ Starlink simply represents their most successful, and most telescopically blinding, campaign.

Starlink

SpaceX's Starlink shocked astronomers with the 2019 launch of 60 internet satellites, appearing as bright streaks in ground-based telescopes.⁶⁰ Now, this number is over 10,000, about two-thirds of the global total.⁶¹ As it stands, the Vera Rubin Observatory, one of the most advanced telescopes in the world, is expected to see Starlink streaks in up to 40% of all images.⁶² Starlink's impact on astronomy is further compounded by the satellite's radio leakage problem, slipping signal into frequencies used by radio astronomers.⁶³

Google's contribution to Starlink cannot be overlooked. Google's 7% stake in SpaceX, purchased in 2015, is now valued at over \$111 billion, with SpaceX itself set to make history as the biggest initial public offering in history, at a \$1.5 trillion valuation.⁶⁴ In 2021, Google secured a 7-year deal with SpaceX to connect Starlink satellites with Google's in-house data network, essentially making Google Starlink's sole cloud computing provider.⁶⁵ The effects of Starlink on astronomy is therefore not attributable to SpaceX and the orbit of Elon Musk alone—Google, too, ought to be considered in the equation.

In collaboration with SpaceX engineers—and spurred by the frustrations of the community—astronomers have studied various efforts to lower Starlink's visibility in the night sky. In a simulation of Starlink's initially modified V1.5 and darker V2 satellites on the Vera

⁵⁹ Brodtkin, "Google and AT&T Invest in Starlink Rival for Satellite-to-Smartphone Service."

⁶⁰ Boyle, "Satellite Constellations Are an Existential Threat for Astronomy."

⁶¹ O'Callaghan, "More than 10,000 SpaceX Starlink Spacecraft Now Orbit Earth—and the Satellite Surge Shows No Sign of Stopping."

⁶² Pultarova, "Satellite Streaks."

⁶³ Grigg et al., "The Growing Impact of Unintended Starlink Broadband Emission on Radio Astronomy in the SKA-Low Frequency Range."

⁶⁴ Ming and Bergman, "How a SpaceX IPO Could Deliver Google One of the Most Lucrative Startup Wins Ever."

⁶⁵ Mohney, "Google Cloud Connects to SpaceX Starlink, Will Host Ground Stations at Data Centers."

Rubin Observatory, Kandula et al. (2025) found that ~99% of satellites were expected to obtain the dimness recommended by the astronomical community in visible wavelengths.⁶⁶ While dimmer V2 satellites in visible wavelengths were corroborated by observations by Mallada (2023),⁶⁷ V1.5 satellites—already above astronomer’s preferred threshold—actually appeared to increase in brightness over time.⁶⁸ Furthermore, Bassa et al. (2024) took direct measurements in radio wavelengths to find a whopping 32x increase in brightness,⁶⁹ and on SKA—which was previously disturbed by Google’s internet balloons—up to 30% of images would be affected.⁷⁰ Suffice to say, there is a great deal of work to be done, and SpaceX’s launches have not appeared to slow. In 2025, SpaceX launched ~3,000 more V2 satellites, totaling over 7,000 of that model in orbit.⁷¹ Indeed, in early 2026 SpaceX filed to launch “one million satellites to power artificial intelligence;” to the naked eye, they would be more numerous than the stars.⁷² SpaceX is moving at a pace foreign to astronomers, practically daring the community to keep up—or risk the night sky as we know it.

AI Data Centers in Space

SpaceX isn’t the only company eyeing low-earth orbit to solve the space, energy, and political⁷³ limitations plaguing contemporary AI data centers. In late 2025, Google announced Project Suncatcher, an ambitious plan to launch solar-powered AI data centers in low-earth orbit by early 2027.⁷⁴ To take on the project, Google will partner with satellite imagery company

⁶⁶ Kandula et al., “Simulated Impact on LSST Data of Starlink V1.5 and V2 Satellites.”

⁶⁷ Mallama et al., “Starlink Generation 2 Mini Satellites.”

⁶⁸ Mallama, “Secular Brightness Trends of Starlink and OneWeb Satellites.”

⁶⁹ Bassa et al., “Bright Unintended Electromagnetic Radiation from Second-Generation Starlink Satellites.”

⁷⁰ Grigg et al., “The Growing Impact of Unintended Starlink Broadband Emission on Radio Astronomy in the SKA-Low Frequency Range.”

⁷¹ “Jonathan’s Space Report | Space Statistics.”

⁷² Tonkin and Massey, “SpaceX and Reflect Orbital Plans Would ‘permanently Scar’ Night Sky.”

⁷³ Lissner, “A Populist Backlash Over AI Is Brewing in America.”

⁷⁴ “Exploring a Space-Based, Scalable AI Infrastructure System Design.”

Planet Labs, another corporation in which Google's parent company, Alphabet, holds the greatest proportion of shares, at over 10%.⁷⁵ In the pre-print paper describing technical aspects of the project, impacts to astronomy is not mentioned once.⁷⁶

Google furthermore appears to take a multi-pronged approach for Project Suncatcher. Perhaps to maintain ties to a launch-capable corporation following SpaceX going public, former Google CEO Eric Schmidt purchased Relativity Space in 2025, with confirmed goals to launch AI datacenters.⁷⁷ Eric Schmidt, coincidentally, is a significant funder to astronomy through the philanthropic fund Schmidt Sciences, which he founded with his wife, Wendy Schmidt. Greater detail on this subject will continue in *Philanthropy*.

The current campaign against AI datacenters in space is spearheaded by the Royal Astronomical Society, The European Southern Observatory, and the International Astronomical Union; all submitted comments to the US Federal Communications Commission (FCC).⁷⁸ Astrobiters, a publication operated by astronomy graduate students, released an article requesting readers to submit public comment to the FCC.⁷⁹ DarkSky International, an advocacy group championing the preservation of dark sky sites, released a statement in opposition to SpaceX space data centers,⁸⁰ and Mongabay, an environmentalist media nonprofit, released an article criticising the proposition.⁸¹ SpaceX, for that matter, any tech corporation with such ambitions, is yet to comment.

⁷⁵ Tickergate, "Planet Labs PBC Ownership | Who Owns Planet Labs PBC?"

⁷⁶ Arcas et al., "Towards a Future Space-Based, Highly Scalable AI Infrastructure System Design."

⁷⁷ Berger, "Eric Schmidt Apparently Bought Relativity Space to Put Data Centers in Orbit."

⁷⁸ Schmidt Sciences, "Home."

⁷⁹ Bonidie and L'Argent, *Two Satellite Proposals Threaten Dark and Quiet Skies Worldwide*.

⁸⁰ *DarkSky International Opposes Reflect Orbital's Proposed Orbital Illumination System*.

⁸¹ Scherer, "Rush to Put AI Data Centers in Space Poses Poorly Understood Dangers."

Opinion

From the litany of Google's commercial interests in space, it appears the trajectory of the space industry is not in astronomer's favor. The precedent has been set: Google, SpaceX, and competing corporations will carry on their ambitions, with or without astronomers, nor, so it seems, a long-term environmental plan. There are no guardrails, and unlike Project West Ford, astronomers cannot present themselves as constituents of Google like they may be of the US government. Indeed, it is perhaps more difficult to contact the Google scientists—among them, astronomers—working on such projects, as Google's initiatives are run out of in-house facilities, not peer universities.

What are astronomers to do? It may be time to adjust tactics beyond government comments. Astronomers could follow suit with the small towns rallying against local data centers,⁸² taking similar tracks as representatives of outer space. Or, follow suit alongside Google's in-house advocacy groups such as No Tech for Apartheid,⁸³ and tech workers' unions.⁸⁴ Above all, I believe it necessary to collaborate with similar interest groups, engaging in one-on-one conversations raising concerns regarding Google's—and all of Big Tech's—capture of science. There is nothing to be gained in stasis, there is the future of humankind to win through action.

Google's Ubiquity in Astronomy

Google's capture of modern work environments must be considered in an analysis of the company's impact on astronomy. Google dominates the search engine market share,⁸⁵ owns the

⁸² NBC News, "Study Shows State and Local Opposition to New Data Centers Is Gaining Steam."

⁸³ <https://www.notechforapartheid.com/>

⁸⁴ Nolan, "Google DeepMind Workers in the U.K. Vote to Unionize over Military AI Contracts amid Internal Backlash over Its Pentagon Deal."

⁸⁵ Statista, "Global Search Engine Market Share 2025."

two most visited websites google.com and youtube.com,⁸⁶ owns gmail.com, the second-most opened email service globally⁸⁷ and most used in the US,⁸⁸ operates Google Drive, the most used file sharing software globally,⁸⁹ and operates a myriad of other services such as Google Calendar, Google Flights, video conferencing software Google Meet, and the AI agent Gemini, among others. These services are used in astronomy much like any other digital work.

There are additional Google services that are especially utilized in astronomy. Google Cloud, the company's data management wing, supports the Vera Rubin Observatory's 500-petabyte dataset.⁹⁰ Google research teams released two crucial open-source softwares for big data computing and ML/AI, JAX and TensorFlow, both used widely in astronomy software. Google Gemini, the company's agentic AI chatbot, is beginning to see use in research applications as well.

The use of Google's suite of services poses a unique complication to the field's relationship with the company. That is, while Google is entrenched in space commercialization initiatives drawing astronomer's ire, astronomers simultaneously take advantage of Google's advanced data storage and ML/AI network—the very same network set to be launched in space! Furthermore, Google's computation tools are frequently used in a military capacity,⁹¹ adding these technologies to the long list of dual-use—military and civilian—innovations that astronomy frequently uses.

⁸⁶ Statista, “Top Websites by Unique Visitors 2026.”

⁸⁷ Statista Daily Data, “Infographic: The World's Most Popular Email Clients.”

⁸⁸ Statista Daily Data, “Infographic: Gmail Dominates U.S. Market.”

⁸⁹ Statista, “Global File Sharing Software Market Share by Vendor 2026.”

⁹⁰ Google Cloud, “Rubin Observatory & SLAC Case Study.”

⁹¹ E.g. US Department of War, “The War Department Unleashes AI on New GenAI.Mil Platform.”

Google Cloud

Cloud services are fundamental to the future of astronomy. The newest generation of telescopes are set to produce one thousand times more data than their predecessors, a scale that mandates cloud data centers.⁹² Such centers allow astronomers to do cloud computing, a process that “bring[s] the code to the data” as opposed to directly downloading unwieldy files to a local network.⁹³ Cloud computing also allows astronomers to parallelize—run multiple programs at once—boosting research speeds.⁹⁴ In the ethos of open-source, cloud-based astronomical data is largely available to the public, limiting the necessity of advanced personal computers.

Google’s cloud computing division, Google Cloud, has initiated a series of partnerships in astronomy. Landoni et al. (2019) described using Google Cloud for statistical modeling, simulations, and data reduction.⁹⁵ Leveraging Google Cloud services, the Asteroid Institute—a nonprofit organization dedicated to protection from asteroid impacts—employed computationally expensive algorithms to identify 104 asteroids in 2022.⁹⁶ Two years later, this project was scaled to 30,000 detections.⁹⁷ Most recently, Google Cloud secured a partnership with the Vera Rubin Observatory to host 500 petabytes of the telescope’s data in a bid to “democratize science.”⁹⁸

Open-source software

In astronomy, Google’s software contributions are primarily through the programs JAX, which accelerates numerical computation, and TensorFlow, a machine learning framework. Both

⁹² Choi, “Migrating Big Astronomy Data to the Cloud.”

⁹³ Shen, “Cloud Computing for (Observational) Astronomy.”

⁹⁴ Ibid.

⁹⁵ Landoni et al., “Application of Google Cloud Platform in Astrophysics.”

⁹⁶ Google Cloud Blog, “How Google Cloud and B612 Discover New Asteroids.”

⁹⁷ Fiaschetti, “Press Release: Asteroid Institute and Google Cloud Identify 27,500 New Asteroids.”

⁹⁸ Google Cloud, “Rubin Observatory & SLAC Case Study.”

are written in Python, the standard programming language in modern astronomy, and released to the public open-source. These softwares occupy distinct niches in ML/AI, thus, I treat these two softwares as different regimes. I compare JAX to Anaconda, Inc.’s computation accelerator

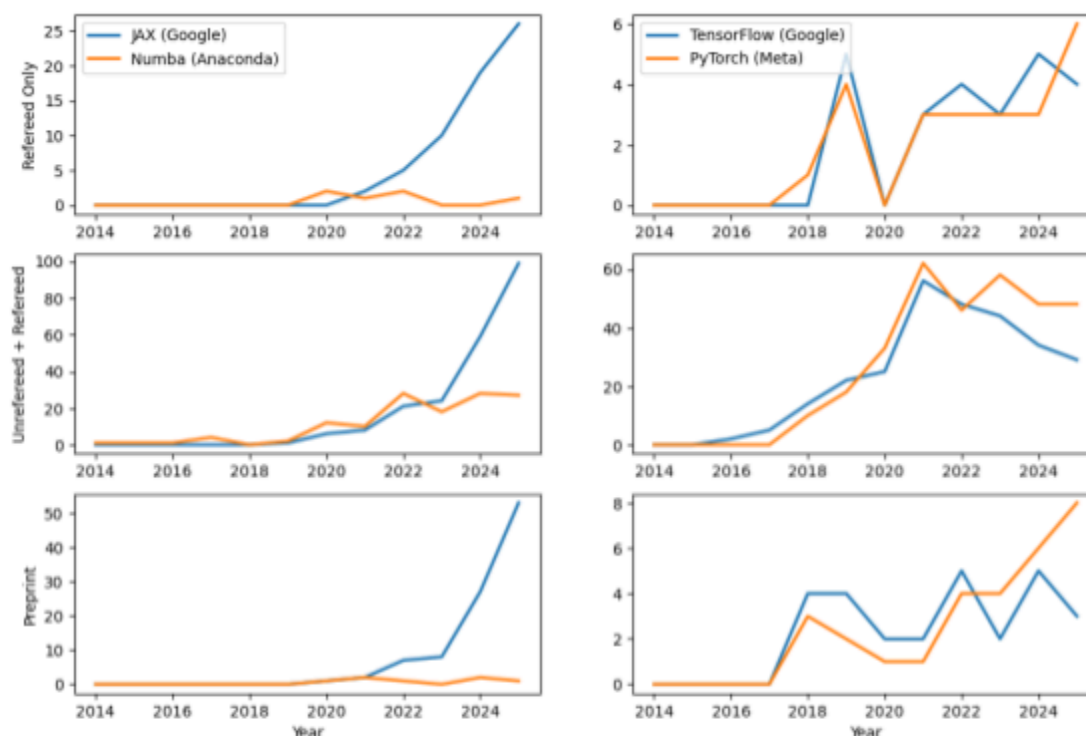


Figure 1. Comparison of Google software use in astronomy papers. “Refereed Only” and “Unrefereed + Refereed” papers are drawn from keyword searches in the SciX catalog (previously NASA/ADS), “Preprint” papers are drawn from the arXiv service.

Numba and TensorFlow to Meta’s machine learning alternative, PyTorch.

The data in *Figure 1* spans astronomy papers, conference abstracts, and preprint materials published between 2014 and 2025. The first row, labeled “Refereed Only,” consists of peer-reviewed papers listed in the SciX catalog. The second row, labeled “Unrefereed + Refereed,” consists of peer-reviewed papers as well as conference papers, typically from AAS meetings. The final row, “Preprint,” consists of all materials released in the arXiv database,

which are typically uploaded before publication or are not sent to publishers at all. The left column traces references to JAX and Numba, and at right, references to TensorFlow and PyTorch. On the left, between JAX and Numba, it is clear that following 2022 JAX is essentially the standard. The right column indicates an essentially neck-and-neck race between Google and Meta for dominance in machine learning software, until Meta apparently wins out sometime between 2022 and 2025. As it appears that Google is transitioning away from TensorFlow in favor of JAX⁹⁹ (and from the plot, astronomers are transitioning to either JAX or PyTorch) I only discuss the JAX program below.

JAX

JAX covers a great deal of functionalities integral to computational, ergo astronomical, research. JAX does two things quite well: use multiple processors at once, and take derivatives. That is, for expensive computational tasks—such as training ML/AI models or big data analysis—JAX allows one to parallelize, splitting the tasks across many processors. Thus, using compute nodes with many processors such as those operated by universities and tech corporations, the program will run much faster. Likewise, in training ML/AI models, optimization—mathematically calculated by taking many, many derivatives—is necessary to improve model accuracy. Such calculations are typically quite slow, but JAX accelerates these processes through its own framework under the hood. These functionalities are further wrapped in a Just-In-Time (JIT) compiler, meaning JAX automatically looks to optimally run its programs in a certain order to maximize speed.

To date, there are ~23 astronomy research projects based on JAX, solving problems related to nearly every subfield. Roughly 2,000 Github users have “starred” (i.e., bookmarked)

⁹⁹ Google Cloud Blog, “6x Faster Migration from TensorFlow to JAX.”

these research projects,¹⁰⁰ and about 300 citations between the projects that are published.¹⁰¹ This count does not include packages that are also widely used but based on JAX in some way, such as PyMC¹⁰² and NumPyro¹⁰³, which are referenced in the literature 411¹⁰⁴ (101 in astronomy)¹⁰⁵ and 540¹⁰⁶ (40 in astronomy) times, respectively. In addition, the top 20 contributors on the JAX team include two astronomers, the most represented field on the team outside of computer science or electrical engineering.¹⁰⁷

Google Gemini

Large language models (LLMs) such as ChatGPT have become pervasive in scientific research. Per a 2026 study surveying researchers in Germany, 43% used LLMs to write code, 18% to analyse data, and 31% to read literature.¹⁰⁸ These generalized tasks are now accompanied with the use of LLMs directly as a research software in and of itself. In October of 2025, a research team at Oxford collaborated with Google’s LLM Gemini to identify transient astronomical events—such as supernovae, the bright explosion of a star’s death—against spurious signals due to noise.¹⁰⁹ The project was the first instance of Gemini’s use as a substitute for bespoke data analysis, and a quite effective one at that—Gemini required only 15 sets of images, far less than the thousands typical of such a project.¹¹⁰ Another study in early 2026 used

¹⁰⁰ JAXtronomy, *JAXtronomy/Awesome-JAXtronomy*.

¹⁰¹ Cite in Appendix A?

¹⁰² PyMC: A Modern and Comprehensive Probabilistic Programming Framework in Python, Abril-Pla O, Andreani V, Carroll C, Dong L, Fonnesbeck CJ, Kochurov M, Kumar R, Lao J, Luhmann CC, Martin OA, Osthege M, Vieira R, Wiecki T, Zinkov R. (2023)

¹⁰³ Phan et al., “Composable Effects for Flexible and Accelerated Probabilistic Programming in NumPyro.”

¹⁰⁴ Abril-Pla et al., “PyMC.”

¹⁰⁵ “‘pymc’ - Science Explorer Search.”

¹⁰⁶ Phan et al., “Composable Effects for Flexible and Accelerated Probabilistic Programming in NumPyro.”

¹⁰⁷ jax-ml, *Jax-Ml/Jax*.

¹⁰⁸ Chugunova et al., “Who Uses AI in Research, and for What?”

¹⁰⁹ Stoppa et al., “Textual Interpretation of Transient Image Classifications from Large Language Models.”

¹¹⁰ Ibid.

Gemini Deep Think—Google’s most advanced rendition of Gemini to date¹¹¹—to calculate the power spectrum of gravitational radiation emitted by cosmic strings, an open problem in theoretical astrophysics.¹¹² Through a series of prompts, Gemini ultimately found six different approaches to solving the problem, three of which result in numerically stable solutions.¹¹³

LLM use in astronomy research has been the subject of concern since their introduction. Zabloudoff et al. (2026) found that LLMs—at least, those less complex than Gemini Deep Think—put towards astronomy research produced logical fallacies, write code that doesn’t work, and hallucinate citations.¹¹⁴ Hogg (2026) details two potential extremes: “let-them-cook,” which would entail releasing the full capabilities of LLMs to conceive of and produce astronomical research, and “ban-and-punish,” outlawing the technology in research entirely.¹¹⁵ It appears that the Google engineers putting Gemini into astronomy research might advocate for the former, and Zabloudoff’s findings would lean towards the latter. Both works identify LLMs as essentially a black box which no one knows the inner workings of, and I wish to further this point: not only are LLMs black boxes, the companies that produce them are as well.

Opinion

I posit that Google’s role in astronomy is akin to some amalgam of industry influence on academia and Google’s consumer monopoly. In the cases above, Google diverges from the corporate tradition in that the company does not primarily employ money towards professorships, research centers, or other material donations. Google already has what astronomers want: near-infinite computational resources and well-documented code to run on it.

¹¹¹ Google DeepMind, “Gemini 3.1 Deep Think.”

¹¹² Brenner et al., “Solving an Open Problem in Theoretical Physics Using AI-Assisted Discovery.”

¹¹³ Ibid.

¹¹⁴ Zabloudoff et al., “Can LLMs Produce Original Astronomy Research in a Semester?”

¹¹⁵ Hogg, “Why Do We Do Astrophysics?”

Thus Google does not necessarily need to pump funds into specific astronomy departments or programs—it simply needs to provide computer resources to astronomers.

Google’s self-positions these initiatives as essentially philanthropic, leaning on how much Google is “helping” astronomers, “democratiz[i]ng science,” and so on.¹¹⁶ Akin to that around Starlink satellites, such phrasing intentionally diverts from the benefits that Google accrues from research use of Google Cloud, JAX, TensorFlow, and Gemini. Google’s return on investment here is both material and ideological.

Alexy et al. (2014) describes two benefits to open-source release of internal programs, stating that “[f]irms significantly improve their performance when (a) opening resources reduces their cost base while (b) strongly increasing demand for their still-proprietary resource(s).”¹¹⁷ The release of Google’s programs open-source allows for volunteer work, where programmers looking to improve the tool for their own use are able to improve it directly, without Google incurring any labor costs. Nearly 1,000 programmers contributed to JAX¹¹⁸ and nearly 4,000 to TensorFlow¹¹⁹ while Google’s DeepMind team is said to consist of ~2,500 engineers.¹²⁰ With both codes already used in-house, hundreds of programmers contributed to Google’s software architecture for free. Google maintains guides on how to pair the paid Google Cloud service with JAX¹²¹ and TensorFlow,¹²² alongside releasing TensorFlow Enterprise to throw in advanced features and long-term support on companies’ Google Cloud purchase.¹²³ Furthermore, the widespread use of Google’s tools also allows the company to influence the broader ML/AI

¹¹⁶ Google Cloud, “Rubin Observatory & SLAC Case Study.”

¹¹⁷ Alexy et al., “Surrendering Control to Gain Advantage.”

¹¹⁸ jax-ml, *Jax-Ml/Jax*.

¹¹⁹ Abadi et al., *TensorFlow, Large-Scale Machine Learning on Heterogeneous Systems*.

¹²⁰ Langley, “Leaked Org Chart Reveals the 58 Top Leaders and Engineers at Google DeepMind as It Races to Surpass OpenAI.”

¹²¹ “A Developer’s Guide to Debugging JAX on Cloud TPUs.”

¹²² Google, “Intermediate ML.”

¹²³ Hackster.io, “Google Launches TensorFlow Enterprise at No Extra Cost for Cloud Customers.”

landscape, as engineers using Google-written code are likely to adopt similar frameworks, thus further ingratiating Google into all of programming work.

Google's contributions to astronomy via cloud computing also adds an interpellative layer to astronomer's perception of the company, forwarding Google's positive image. Google offers its data centers to the very same observatory that Google-funded telescopes are blocking the night sky from, stymieing a wholly negative perception of the company. This ploy has precedent in fossil fuels companies' investments in university environmental research to "green" fossil fuels production, thereby washing their image into one more appealing to the climate conscious.¹²⁴ I posit that Google's implementation of ML/AI—and by extension, cloud computing—in astronomy and the sciences writ large plays a similar role, an attempt to divert eyes from those same tools being used to surveil and sell human attention, unpopular US military applications, and environmental degradation.

Philanthropy

As the following section describes Google's donations to astronomy, this precursor will briefly outline the financial landscape facing the modern university. Since the 1980s, the overarching motive force in the modern university is *corporatization*, where industry-academy partnerships are encouraged—professors make start-ups based on their research, and corporate leaders sit on university boards—thus the *corporate university* is born.¹²⁵ The corporate university is characterized by productivity measures, competition between departments for funds, substituting tenure track positions with part-time instructors,¹²⁶ and rising tuition costs,¹²⁷ among other markers. Researchers at the corporate university are motivated to turn towards

¹²⁴ Graham, "Fossil Knowledge Networks."

¹²⁵ Brownlee, J, *Academia, Inc.: How corporatization is transforming Canadian universities*.

¹²⁶ Ohmann, "Citizenship and Literary Work: Thoughts Without a Conclusion."

¹²⁷ Williams, "Debt Education: Bad for the Young, Bad for America."

corporations, performing research on their behalf in exchange for crucial funding. In so doing, faculty become direct pressure targets for corporations, the latter aiming to suppress findings from the former that might hurt their bottom line.

Google in the Academy

Google's philanthropic funding of academics always seems to further its commercial interests. This is particularly true for professors in law or public policy—the professors traditionally closest to policymakers. Between 2009 and 2017, Google funded about 100 research papers supporting the company's policy interests.¹²⁸ Such funding is rarely reported in the published articles, despite, in many cases, Google staff themselves reviewing manuscripts before publication.¹²⁹ The conditions of such funding are consistently presented as no strings attached, in one case representatives from Google telling a law professor: “If you take this \$20,000 and open up a doughnut shop with it—we'll never give you any more money—but that's fine.”¹³⁰ Thus, in most cases, there is a plausible deniability around Google's direct influence on academics. The funds are simply gifts in the order of thousands to tens of thousands of dollars, a reward for professors with certain claims that align with Google's goals in patent law (e.g. the case of Professor Jorge Contreras),¹³¹ anti-trust policy (e.g. an offer to Professor Daniel Crane),¹³² and data privacy (e.g. the case of Professor Daniel Sokol).¹³³

Funding furthermore comes towards conferences. Such was the case during two online conferences in 2013, both on patent law and neither openly advertised as Google-supported events. Organized by University of Florida law professor Daniel Sokol in communication with

¹²⁸ Mullins and Nicas, “Paying Professors: Inside Google's Academic Influence Campaign.”

¹²⁹ Ibid.

¹³⁰ Ibid.

¹³¹ Ibid.

¹³² Ibid.

¹³³ https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2723693

Google's lawyers, both meetings had Google identify certain professors to Sokol for invitation, as well as curate the topics for their papers. Google paid Sokol \$5,000 for organizing each conference.¹³⁴ In another especially hands-on example, Google funded and essentially co-organized a 2012 meeting at George Mason University's (GMU) Law and Economics Center regarding internet search competition, again without disclosing their involvement. This meeting was scheduled during the ongoing antitrust case by the FTC against Google on the very subject. Speakers were hand-picked by Google, who presented antiregulation arguments to relevant government officials, who were also invited at Google's request. Two other conferences at GMU were organized to similar effect.¹³⁵

Google has also demonstrated its power to outright fire dissidents at research institutions it funds. By 2017, New America—a public policy think tank in Washington, D.C.—had received over \$21 million in funding from Google's then-CEO, Eric Schmidt. That June, Open Markets, a research team at New America, released a statement in support of the European Union's \$2.7 billion antitrust fine against Google.¹³⁶ Schmidt, who had also previously chaired New America until 2016, made his displeasure at such a statement immediately apparent through a comment to Anne-Marie Slaughter, president of New America.¹³⁷ Days later, Slaughter accused Barry Lynn, director of Open Markets, of “imperiling the institution as a whole,” and relieved Lynn and the 10-person Open Markets team from their roles at New America.¹³⁸ While Google's direct role in jettisoning Open Markets from New America remains unclear, the company's soft power within such research institutions should not be understated. Indeed, regardless of the extent that Google micromanages these organizations, the result is clear. In Lynn's words: “Google is very

¹³⁴ Ibid.

¹³⁵ Hamburger and Gold, “Google, Once Disdainful of Lobbying, Now a Master of Washington Influence.”

¹³⁶ Lynn, “Open Markets Applauds the European Commission's Finding Against Google for Abuse of Dominance.”

¹³⁷ Vogel, “Google Critic Ousted From Think Tank Funded by the Tech Giant.”

¹³⁸ Ibid.

aggressive in throwing its money around Washington and Brussels, and then pulling the strings.... People are so afraid of Google now.” Such a climate is further seen in Slaughter’s previous concerns on Lynn’s Google critiques. Prior to a 2016 conference organized by Open Markets warning of monopolisation in tech, Slaughter wrote to Lynn: “We are in the process of trying to expand our relationship with Google on some absolutely key points... just THINK about how you are imperiling funding for others.”¹³⁹

Thus, from Google’s history in lobbying academics, it is unsurprising that the company regularly donates hundreds of millions to universities. A \$1 billion donation to AI research at over 100 universities,¹⁴⁰ while a tool to train more data scientists for Google, is simultaneously a bargaining chip that Google has been more than willing to use to ensure research reaches their aims, as in the case of New America. Indeed, with the precedent laid out by Google’s influence on public policy academics, such a process may be translated to other fields, or scaled up to the level of a university.

Philanthropic Endeavors in Astronomy

While there have yet to be recorded instances of direct research suppression in astronomy due to corporate influence, there are two notable examples of Google’s funding with the potential of such an event. These cases vary significantly in magnitude: a \$1 million donation to ensure that Lick Observatory remains open,¹⁴¹ and an estimated \$500 million investment by ex-Google CEO Eric Schmidt to construct a 4-telescope observatory system.¹⁴²

¹³⁹ Ibid.

¹⁴⁰ Cai, “Google Commits \$1 Billion for AI Training at US Universities.”

¹⁴¹ SpaceRef, “Google Gives Lick Observatory \$1 Million.”

¹⁴² Berger, “Former Google CEO Plans to Singlehandedly Fund a Hubble Telescope Replacement.”

Lick Observatory

In June 2013, the University of California Observatories (UCO) Board recommended to the UC Office of the President (UCOP) to defund Lick Observatory in favor of the newer, larger Keck and Thirty Meter telescopes in Hawaii. Soon after, UCOP announced that Lick's \$1.5 million annual budget would decrease to \$1.3 million in 2016, and take a "glide path" to zero in 2018, a response to UC's research budget decreasing by 40% since 2007.¹⁴³ Lick's budget was already down from \$2.5 million prior to the 2008 recession.¹⁴⁴ Astronomers pointed towards Steven Beckwith, former UC vice president of research and graduate studies, as the primary motivator for the cut, having "publicly belittled the merits of ground-based telescopes" which are hosted at Lick. Beckwith, in turn, motioned that the UCO Board—which Beckwith convened and appointed each member of in 2012—were responsible for the decision. In turn, the UCO Board deferred to the UC administration, stating: "ask the systemwide provost about this.... I will not comment." The systemwide provost, Aimee Dorr, swung back to the board, vaguely stating that the decision "was based on all of these constraints and all this advice [from the UCO Board]."¹⁴⁵

This lack of transparency seems better fit for a large company than a university; it is representative of the supersized bureaucracy emblematic of corporate universities. Indeed, the UCO Board was largely composed of "budget-minded administrators," meaning 7 out of 13 members had no astronomy experience at all. Garth Illingworth, an astronomy professor at UC Santa Cruz, called out such a corporate framework explicitly, stating that Beckwith sees himself "like a CEO in a company," and that the UCO Board was "just a bunch of 'yes' people... not

¹⁴³ Hoban, "Facing a Waning Future."

¹⁴⁴ Cal Alumni Association, "Lick Gets Googled—But Is Cool Million Enough to Save the Endangered Observatory?"

¹⁴⁵ Hoban, "Facing a Waning Future."

very interested in Lick or UC astronomy.”¹⁴⁶ In the face of a backsliding budget, the professional administrators made the call to ditch Lick.

After over a year of faculty, congressional,¹⁴⁷ and student pressure,¹⁴⁸ UC administration ultimately reinstates Lick’s \$1.5 million annual funding.¹⁴⁹ But the root problem remains: the budget—limited by the recession and decreases in state funding¹⁵⁰—is still “spartan,” still well below pre-recession levels. It is here that Google steps in. Following the UC administration’s recommendation that astronomers look to Silicon Valley for donations and some inquiry, Google takes up the mantle with a \$1 million donation over two years—nearly returning Lick to pre-recession levels—starting in early 2015.¹⁵¹ Soon after, enough philanthropic donations came in to support Lick sustainability, at least for the near future. Google’s donation naturally did wonders for the company’s public relations,¹⁵² and plausibly represents an intentional shift of the public focus away from the significant antitrust suit settled less than 3 years before.¹⁵³

However, Google had additional interests in space preceding its donation to Lick Observatory. In 2014, Google had purchased Skybox Imaging, a company that produced small satellites with high-resolution cameras. Partnering with Fidelity Investments, the pair purchased a combined \$1 billion investment in Space Exploration Technologies, better known today as

¹⁴⁶ Ibid.

¹⁴⁷ Ibid.

¹⁴⁸

https://www.dailycal.org/archives/save-lick-observatory-to-defend-public-education-scientific-inquiry/article_fde8d265-1773-55a5-8891-3f1da90f162f.html

¹⁴⁹

https://www.dailycal.org/archives/uc-overturns-decision-to-withdraw-all-funding-to-lick-observatory/article_819e1693-b75d-5b97-bd16-48087186675b.html

¹⁵⁰

https://www.dailycal.org/archives/uc-does-not-plan-to-shut-down-lick-observatory/article_a5ba90da-3cb7-542e-80a3-0c93d2f427f8.html

¹⁵¹ Cal Alumni Association, “Lick Gets Googled—But Is Cool Million Enough to Save the Endangered Observatory?”

¹⁵² Betz, “Google Helps Lick Observatory Stay Open with \$1 Million Gift.”

¹⁵³ *Bloomberg.Com*, “Google Judge Accepts \$22.5 Million FTC Privacy Settlement.”

SpaceX, with the dream of satellite internet access.¹⁵⁴ While it is unclear whether Google intentionally prioritized donations to Lick Observatory to deflect potential scrutiny for contemporaneous investments in satellites, low-Earth orbit satellites such as those used by Skybox and SpaceX, have been a concern for astronomers for some time. The launch and destruction of these satellites can produce debris that orbits the Earth near-indefinitely, essentially a hail of particles moving fast enough to destroy another satellite, and, in so doing, producing more debris. The International Space Station notably had to maneuver around space junk three times in 2014,¹⁵⁵ and recommendations to limit the number of satellites in low-Earth orbit have been announced multiple times by the United Nations.¹⁵⁶ In any case, Google found synergy with Lick Observatory, and its donation won the company brownie points for astronomers for the duration of its contributions, if not longer.

Schmidt Observatories

The recently announced Schmidt Observatories, entirely funded by ex-Google CEO Eric Schmidt, raises additional concerns regarding the potential of external influence in astronomical research. At a 2026 American Astronomical Society (AAS) meeting, Schmidt Sciences—a philanthropic project funded by Eric and Wendy Schmidt—unveiled plans to construct an entirely private observatory system, set to go online as early as 2027.¹⁵⁷ The system will comprise four telescopes: the Argus Array, the Deep Synoptic Array (DSA), the Large Fiber Array Spectroscopic Telescope (LFAST), and the Lazuli Space Observatory.¹⁵⁸ Each

¹⁵⁴

<https://www.nytimes.com/2015/01/22/business/google-hopes-to-take-the-web-directly-to-billions-lacking-access.html>

¹⁵⁵ *BBC News*, “Trouble in Orbit.”

¹⁵⁶ https://www.unoosa.org/pdf/publications/st_space_49E.pdf

¹⁵⁷

<https://www.scientificamerican.com/article/schmidt-sciences-announces-plan-for-lazuli-a-private-space-telescope/>

¹⁵⁸ <https://www.schmidtsciences.org/schmidt-observatory-system/>

ground-based project is under development contracts with universities: Argus with the University of North Carolina, Chapel Hill,¹⁵⁹ DSA with the California Institute of Technology,¹⁶⁰ and LFAST with the University of Arizona.¹⁶¹ Schmit Sciences will develop Lazuli in house.^{162,163} Data from the Schmidt Observatory system is expected to be publicly available in regular updates, a similar model to the publicly-available datasets through NASA.

Schmidt Sciences, however, operates on a different funding model than traditional government contracts. In the words of Dr. Feldman, Schmidt's chief scientist, "either we do it or we don't, and the Schmidts either choose to give us the appropriate amount of money or not."¹⁶⁴ This may perhaps raise flags for those who recall the earlier story of Barry Lynn—Schmidt has made it clear that if scholars oppose the priorities of Google, he has no problem cutting off funds.

Not long after the announcement of Schmidt Observatories, Schmidt published an op-ed advocating for greater philanthropic investment in science, with visions of "private missions to the moon and Mars," accelerated by a "commercial mindset [to] scale faster, take more risks, drive down costs."¹⁶⁵ Simultaneously, Schmidt presses concerns regarding China's space program, particularly in lunar exploration. Recalling Cold War rhetoric, Schmidt worries that the US "faces a real threat of losing the competition to establish a permanent presence on the Moon."¹⁶⁶ Funnily enough, the rhetoric to beat China is also employed by fossil fuels companies¹⁶⁷—global warming doesn't matter, only maintaining American dominance does.

¹⁵⁹ <https://argus.unc.edu/>

¹⁶⁰ <https://www.deepsynoptic.org/>

¹⁶¹ <https://astro.arizona.edu/research-groups/lfast>

¹⁶² <https://www.schmidtsciences.org/schmidt-observatory-system/#modal-lazuli-space-observatory>

¹⁶³ <https://arxiv.org/abs/2601.02556>

¹⁶⁴ <https://www.nytimes.com/2026/01/09/science/eric-schmidt-telescopes-google-space.html>

¹⁶⁵ <https://thehill.com/opinion/technology/5699050-space-exploration-public-private/>

¹⁶⁶ Ibid.

¹⁶⁷ Gustafson 2012 pg 311

Opinion

We are not yet at the stage in which we might find a Barry Lynn case for astronomers. However, I caution that such a reality may come to pass. The maneuvers of Google and Schmidt indicate that there is more than meets the eye with their philanthropy, and should astronomers miss that, a rug of funding might sweep out from under them if the wrong thing is said.

Schmidt's simultaneous funding of an observatory system named after him which will run research programs under his control, plans to place data centers into space with the launch capability of Relativity Space, where he is the CEO, and history of loyalty to Google, including ousting scholarly critics, represents a system of control in astronomy that is growing ever tighter. Add atop that Schmidt's past interests in asteroid mining,¹⁶⁸ and it appears that Schmidt's "purely charitable"¹⁶⁹ initiatives might have additional motives. Indeed, Schmidt's leadership at Google—which afforded him at least \$100 million in equity¹⁷⁰—stretched from 2001 to 2020,¹⁷¹ a timeframe that included the company's search engine dominance and heavy investments in SpaceX. Astronomers' frustrations with Google-funded Starlink may yet find a direct avenue of expression—it appears that astronomy itself will be Google-funded as well.

Conclusions

I posit that Google's primary objective is to profit from the space industry, while the company's ubiquity and philanthropy are forms of interpellation. Besides the possibly coincidental donation to Lick Observatory amidst Google's SpaceX investment, all of Google's involvements in astronomy follow the first launch of SpaceX's starlink satellites in 2019.

¹⁶⁸ <https://www.bbc.com/news/science-environment-17827347>

¹⁶⁹ <https://astro.arizona.edu/news/eric-and-wendy-schmidt-fund-space-telescope-three-ground-based-observatories>

¹⁷⁰ <https://www.reuters.com/article/idUSTRE70M1V120110123/>

¹⁷¹

<https://www.cnet.com/tech/tech-industry/eric-schmidt-who-led-googles-transformation-into-a-tech-giant-has-left-the-company/>

Immediately after, mass outrage erupted in the astronomy community as the satellites contributed to light pollution and blocked out telescope data. In the years since, SpaceX has yet to dim its satellites to the requested levels, despite active engagement with the IAU and European Southern Observatory. Satellites continue to launch with increasingly fantastical plans, and astronomers continue voicing their frustrations.

In order to nullify these voices, Google turned to the strategies set by fossil fuel companies—ingrain yourself into the field until you are synonymous with research within it, making yourself *ubiquitous*. However, Google does not need to follow the exact same path as its fossil fuel forebears in purchasing office buildings and endowing professorships. Indeed, Google already has the most valuable resource for an astronomer—fast, scalable computation programs and near-unlimited data storage capacity. The open-source release of JAX and TensorFlow, a move that pushed Google-based code to the forefront of modern computing anyways, as well as data partnerships with the Vera Rubin Observatory and university astronomy teams directly, interpellate astronomers as, in essence, Google customers. Astronomers thus get first-hand confirmation of a notion already popular in scientific fields: that technological progress (in the particular, extractive ways it comes about today) is a given, and for every technology you must take the bad in with the good. You must accept the Starlink satellite streaks—let alone privacy invasion, environmental degradation, and military contracts—alongside the massive resources Google affords you.

That is not to say that Google is absent of any financial support. The Schmidt Observatory System is a magnum opus in endowed research facilities. Nevermind Schmidt's history of sacking academics that disagreed with him, or stated intentions to launch Google data centers to space with the rocket company he is now CEO of. In such a volatile funding structure,

Schmidt seems to say, perhaps it is better not to bite the hand offering to feed you. Even if it means ground-based telescopes will lose massive fractions of data, and there may not be a dark sky zone left in a couple decades.

I ask of astronomers to remain critical thinkers in uncertain times. Not necessarily to boycott funds from questionable sources—funding is funding, after all. But, to be cognizant that the multinational corporations seeking to colonize space are not likely to be giving out resources for free, be they monetary or computational. It may yet be possible, should the astronomy community effectively organize themselves, to pressure such corporations and the federal government strongly towards an outer space that is truly reserved for the benefit of all humankind. Do not let the incoming gifts from space-profiteering parties sway you from that goal.

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